35 · Hormonal control of sperm, and testosterone

The words of this track, as it says them. Bold brown is the examiner.

The hormonal control of spermatogenesis, and testosterone. Five marks.

Describe the hormonal control of spermatogenesis, and the physiological effects of testosterone during puberty.

The hook: GnRH, then FSH and LH. L for Leydig, T for testosterone. S for Sertoli, I for inhibin.

From the brain. The hypothalamus releases GnRH, gonadotropin-releasing hormone, which makes the anterior pituitary release FSH and LH.

At the testes. LH stimulates the Leydig cells, also called interstitial cells, to make testosterone, a lipid-soluble hormone made from cholesterol. Testosterone enhances spermatogenesis. FSH acts on the Sertoli cells, also called sustentacular cells, which nourish the developing sperm and support spermatogenesis.

Negative feedback. Testosterone switches off the hypothalamus, so there's less GnRH, and the anterior pituitary, so there's less LH. Inhibin, from the Sertoli cells, switches off FSH. That keeps the sperm count steady.

Testosterone at puberty. It grows the reproductive organs to adult size, and brings the secondary sex characteristics: pubic, axillary and facial hair, a deeper voice, a bigger chest, and thicker, oilier skin. It increases bone density and skeletal muscle mass. And it's the basis of libido, in both males and females.

And in females? Her student-marked quiz asks for testosterone in males and females, and her model answer covers both. Testosterone has lots of effects around the body, and we see them in females too, because the adrenal glands can make testosterone. In men, it's thought to regulate sex drive, or libido, bone mass, fat distribution, muscle mass and strength, and the production of red blood cells and sperm.

The traps. Leydig cells make testosterone, driven by LH. Sertoli cells nourish the sperm and make inhibin, driven by FSH. Don't swap them. Second: inhibin controls FSH, while testosterone controls GnRH and LH. Third: give the chain in order. Hypothalamus, pituitary, testes.

Once more. The male hormones.

The hypothalamus releases GnRH, so the anterior pituitary releases FSH and LH. LH stimulates the Leydig cells to make testosterone, which enhances spermatogenesis. FSH acts on the Sertoli cells, which nourish the sperm. Testosterone feeds back to switch off GnRH and LH, and inhibin from the Sertoli cells switches off FSH. At puberty, testosterone grows the reproductive organs, and brings pubic, axillary and facial hair, a deeper voice, more bone density and muscle mass, and libido. Females make testosterone too, in the

adrenal glands, and it drives libido in both sexes.